

Notes: Sweeting (Econometrica, 2013)

Dynamic Product Positioning in Differentiated Product Markets: The Effect of Fees for Musical Performance Rights on the Commercial Radio Industry

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1 Big picture

- **Question.** How would radio stations reposition their programming if music stations were required to pay performance-rights fees for recorded music?
- **Policy setting.** The motivating policy is the proposed Performance Rights Act, which would require music radio stations to pay fees to owners of sound-recording performance rights in addition to existing composition-rights fees.
- **Why the question is dynamic.** Stations can switch programming formats, but switching is costly and risky. Therefore, the response to a new fee is not an instantaneous static reallocation; it is a gradual transition in market structure.
- **Industry setting.** The application is commercial radio. Products are differentiated by programming format, stations differ in observable and unobservable quality, advertisers value listener demographics differently, and firms can own multiple stations in the same market.
- **Core model.** The paper estimates a dynamic oligopoly model of format choice. Firms choose whether to keep or change station formats, anticipating future revenues, repositioning costs, fixed costs, economies of scope, and rivals' format choices.
- **Computational challenge.** The model has many stations, multiple formats, multi-station firms, market-level state variables, and continuous unobserved station quality. The paper solves the model using parametric approximations to value functions.
- **Main empirical finding.** High performance fees would reduce the number of music stations significantly and relatively quickly, but not eliminate music programming.
- **Key quantitative result.** A fee equal to 10% of music-station revenues reduces the number of music stations by about 4.6% within two years and by about 9.4% in the long run.
- **Why the effect is limited.** Many listeners—especially advertiser-valued demographics—strongly prefer music programming. Switching away from music can lower a station's ability to attract valuable listeners.
- **Takeaway.** Product variety effects of policy can be large, but credible prediction requires dynamic differentiated-products models rather than static demand estimates alone.

2 Incremental contribution of the paper

2.1 Contribution relative to static differentiated-product models

- Static product-choice models can predict how firms would choose products if adjustment were costless or instantaneous.
- This paper studies a setting where changing product characteristics—radio formats—requires costly repositioning and where firms anticipate future payoffs.
- The incremental contribution is therefore to embed product differentiation inside a dynamic oligopoly model that can trace both short-run and long-run variety responses to policy.

Incremental contribution: The paper moves the empirical analysis of product variety from a static “which products are profitable today” question to a dynamic “how will firms transition after a policy shock” question.

2.2 Contribution relative to previous radio-market studies

- Previous work on radio markets often studied mergers, advertising, or static demand.

- Sweetening studies format repositioning itself, allowing firms to change product types over time.
- The model allows for horizontal differentiation across programming formats, vertical quality differences across stations, demographic heterogeneity in listener preferences, and multi-station ownership.

Incremental contribution: The paper uses the radio industry as a tractable but rich setting for studying dynamic product positioning in differentiated-product markets.

2.3 Contribution relative to dynamic oligopoly estimation

- Dynamic oligopoly models often face a state-space problem when products are differentiated and firms have many possible choices.
- The paper uses parametric approximations to firm value functions, building on conditional-choice-probability methods and value-function approximation ideas.
- It compares estimates based on modified pseudo-likelihood, modified moment, iterated pseudo-likelihood, and forward-simulation approaches.
- It checks whether estimates based on value-function approximations lie within bounds implied by forward simulation and moment inequalities.

Incremental contribution: The paper is methodological as well as applied: it shows how to estimate and solve a rich product-positioning game that would otherwise be computationally infeasible.

2.4 Contribution to policy analysis

- The proposed performance-rights fee had not actually been imposed on broadcast radio, so reduced-form evidence alone could not identify the counterfactual.
- The paper uses a structural model to simulate policy effects under 10% and 20% revenue fees.
- It predicts both the number of music stations and listening shares over transition horizons up to twenty years.

Incremental contribution: The paper provides an early, detailed example of using structural dynamic product-choice models for regulatory policy analysis in media markets.

3 Data and measurement

3.1 Market sample

- The estimation sample contains 102 local radio markets from 2002 to 2005.
- Data come primarily from BIAfn’s MediaAccess Pro database.
- The ten largest Arbitron markets are dropped to reduce computational burden.
- Markets where more than 6% of listening is to stations based outside the market are excluded to avoid modelling cross-market interactions.
- Each market is treated as independent; common ownership across markets is ignored in the structural model.

Interpretation: The sample restriction is a modelling choice. The paper sacrifices some breadth to obtain markets where local strategic interaction can be modelled without explicitly solving a national radio-company game.

3.2 Station sample and timing

- The paper uses Arbitron estimates of station audience shares for people aged 12 and over.
- The data are observed in two quarters per year, Spring and Fall.
- The dynamic model therefore has six-month periods, from Spring 2002 to Spring 2005.
- Noncommercial stations are excluded because they do not compete for advertising revenue in the same way.
- Stations that fail to meet Arbitron’s minimum reporting standard in one or more quarters are excluded.
- Simulcast AM partners of FM stations are dropped from the main sample.
- The final sample contains 2,375 stations and 16,481 station-period observations.

Interpretation: A six-month period is short enough to capture switching dynamics but long enough that a station’s format change can plausibly be observed as a discrete strategic choice.

3.3 Demographic groups and listening shares

- Each market’s population is divided into 18 demographic groups: three age categories, three race/ethnicity categories, and gender.
- Age categories are 12–24, 25–49, and 50 plus.
- Race/ethnicity categories are white, black, and Hispanic.
- Listener demand allows preferences over programming formats to vary across these demographic groups.
- Audience shares are converted into demand quantities using the assumption that every person in the market can listen to radio up to 6 hours per day.

Interpretation: Demographic heterogeneity is crucial for the counterfactual because advertisers value some listener groups more than others, and different groups prefer different formats.

3.4 Format aggregation

- BIAfn lists detailed primary programming formats for each station.
- The paper aggregates detailed formats into seven active categories plus a nonactive “Dark” format.
- The active categories are AC/CHR, Rock/AOR/Classic Rock, Country, Urban, News/Talk, Spanish-language, and Other Programming.
- The first four categories are treated as contemporary music formats in the counterfactual.
- News/Talk, Spanish-language, and Other Programming are treated as nonmusic formats for the performance-fee simulation.

Interpretation: The aggregation keeps the dynamic model computationally feasible while preserving the main economic distinction: formats that would pay performance fees versus formats that would not.

3.5 Station revenues

- BIAfn reports estimated annual advertising revenues for most station-years.
- Revenues are not modelled from advertiser primitives; instead, the paper estimates a reduced-form revenue function.

- Revenues depend on predicted listeners by demographic group and station/market characteristics.
- Market-year fixed effects absorb market-level advertising conditions.

Interpretation: The reduced-form revenue function is sufficient for the counterfactual because the policy changes programming incentives through station revenues rather than through explicit advertiser pricing rules.

4 Model

4.1 Overview of the dynamic game

- In each market m , radio station owners play an infinite-horizon discrete-time game.
- Firms own one or more stations and choose formats for their stations.
- The state includes station formats, observable station characteristics, unobserved station quality, market demographics, and firm ownership.
- Markets are assumed independent.
- Firms use stationary Markov Perfect strategies.

Interpretation: The model treats format choice as a strategic investment decision: firms compare current revenues with future value and the cost of repositioning.

4.2 Timing within each period

Within each six-month period:

1. Firms observe the current state.
2. Firms pay fixed costs for active stations.
3. Firms observe private payoff shocks for their possible format choices.
4. Firms choose formats and receive advertising revenues.
5. The state evolves to the next period through format changes, quality evolution, and demographic changes.

Interpretation: Private shocks rationalize idiosyncratic format decisions that are observed by the firm but not by the econometrician.

4.3 Listener demand

Listener utility from station s for a consumer in demographic group d has a differentiated-products structure with station quality, format effects, and demographic-format interactions. The paper estimates listener tastes for formats and uses these estimates to predict how many listeners in each demographic group each station attracts.

- Format preferences vary sharply across age, gender, race, and ethnicity.
- Older and male listeners prefer News/Talk and some nonmusic formats.
- Black and Hispanic listeners have strong relative preferences for Urban and Spanish-language programming.
- Contemporary music formats tend to attract demographics that are valued by advertisers.

Interpretation: Demand estimates create the key tradeoff in the policy simulation: a station can avoid fees by switching to nonmusic, but it may lose valuable listeners.

4.4 Revenue function

The advertising revenue per listener with demographic D_d is specified as

$$r_A(D_d) = y_m(1 + Y_{sty}\gamma^Y)\alpha + D_d\gamma^D, \quad (1)$$

where y_m is a market-year fixed effect, Y_{sty} contains station characteristics, and D_d is the demographic vector.

- Total station revenue is the sum over demographic groups of predicted listeners times revenue per listener.
- A station's revenue may be affected by format switching and by the number of own or rival stations in the same format.
- In the counterfactual, performance fees are applied as a percentage of music-station revenue.

Interpretation: Advertiser valuation creates a vertical dimension to demand: not all listeners have the same revenue value to stations.

4.5 Repositioning costs

The firm pays a repositioning cost when it changes a station's format. Repositioning costs can vary with:

- market size;
- whether the move is active-to-active, dark-to-active, or active-to-dark;
- the revenue of the switching station;
- whether the station recently switched formats.

Interpretation: Repositioning costs determine the speed of adjustment after performance fees are introduced. High costs slow the transition away from music formats.

4.6 Economies of scope

The paper allows for economies of scope when the same owner operates multiple stations in the same format in a market:

$$FixedCost_s = FixedCost_0 - \theta_C \cdot \mathbf{1}\{\text{sister station in same format}\}. \quad (2)$$

Interpretation: Scope economies matter because a multi-station owner may prefer clustering formats to reduce fixed costs, but clustering can also cannibalize audiences.

4.7 Unobserved station quality

Unobserved station quality evolves according to

$$\xi_{s,t+1} = \rho_\xi \xi_{s,t} + \nu_{s,t+1}, \quad \text{if the station does not switch format,} \quad (3)$$

$$\xi_{s,t+1} = \rho_\xi \xi_{s,t} + \nu_{s,t+1} + \gamma^\xi, \quad \text{if the station switches format.} \quad (4)$$

- ρ_ξ measures persistence in unobserved station quality.
- γ^ξ captures the average quality shift associated with format switching.
- The innovation ν is assumed independent of the firm's current format choice.

Interpretation: The model allows stations to have persistent appeal beyond their format, and it tests whether switching formats causes a quality penalty.

4.8 Bellman equation and choice probabilities

The firm value function satisfies

$$V_o^\Gamma(M_{j,o,t}, \varepsilon_{o,t}) = \max_{a \in A_o(M_{j,o,t})} \{ \pi(a, M_{j,o,t}) + \theta_\varepsilon \varepsilon_{o,t}(a) + \beta E[V_o^\Gamma(M_{h,o,t+1}) \mid a, \Gamma_{-o}] \}. \quad (5)$$

With Type-I extreme value shocks, conditional choice probabilities take logit form:

$$P_o^\Gamma(a \mid M_{j,o,t}) = \frac{\exp(v_o^\Gamma(a, M_{j,o,t})/\theta_\varepsilon)}{\sum_{a' \in A_o(M_{j,o,t})} \exp(v_o^\Gamma(a', M_{j,o,t})/\theta_\varepsilon)}. \quad (6)$$

Interpretation: The model is a dynamic discrete-choice game. The key computational burden is approximating expected continuation values for many markets, firms, formats, and stations.

5 Solution and estimation methods

5.1 Value-function approximation

- Exact solution is infeasible because the state space is too large.
- The paper approximates firm value functions using parametric basis functions.
- The approximation summarizes the state in variables such as station qualities, format counts, ownership, and market characteristics.
- The paper estimates dynamic parameters using methods related to Aguirregabiria and Mira and Pesendorfer–Schmidt-Dengler style approaches.

Interpretation: The value-function approximation is what makes a rich differentiated-products dynamic game computationally tractable.

5.2 Modified pseudo-likelihood and modified moments

- The modified pseudo-likelihood procedure updates conditional choice probabilities and value-function approximations iteratively.
- The modified moments approach uses moments based on observed switching and format choices rather than the full likelihood.
- The paper compares estimates across methods to check robustness.

Interpretation: Consistency across estimation methods increases confidence that results are not artifacts of a single approximation scheme.

5.3 Forward-simulation checks

- The paper also computes estimates based on forward-simulation approximations.
- Moment-inequality bounds provide a check on the plausibility of estimates from parametric approximations.
- The preferred estimates are based on parametric approximations, but the forward-simulation exercise provides external validation.

Interpretation: This is an important methodological safeguard because dynamic oligopoly estimation can be sensitive to value-function approximation errors.

6 Identification

6.1 Demand-side identification

- Format taste parameters are identified by variation in audience shares across markets with different demographic compositions.
- Station quality is identified from persistent differences in station shares after controlling for format and demographics.
- AM/FM band, signal coverage, and out-of-market status help explain station-level quality differences.

Interpretation: Demographic variation is key: if Hispanic-heavy markets have more Spanish listening and black-heavy markets have more Urban listening, the model can infer demographic preferences.

6.2 Revenue-side identification

- Revenue parameters are identified by linking predicted listeners by demographic group to station advertising revenues.
- Market-year fixed effects absorb broad advertising-market conditions.
- Demographic coefficients identify which listeners advertisers value more.

Interpretation: The counterfactual depends critically on the fact that not all audiences generate the same advertising revenue.

6.3 Dynamic-cost identification

- Repositioning costs are identified from observed rates of format switching.
- Differences in switching rates across market sizes identify market-size-specific costs.
- Moves to and from Dark identify active/dark transition costs.
- Multi-station ownership choices identify economies of scope.
- The scale of payoff shocks is identified because revenues are treated as observed when estimating the dynamic model.

Interpretation: Low observed switching rates imply either high repositioning costs or large private payoff shocks. The model uses switching patterns, revenues, and market structure to separate these forces.

6.4 Limits of identification

- The paper treats entry and exit as unanticipated shocks rather than endogenous choices.
- It does not structurally model advertising demand.
- It ignores cross-market ownership economies of scope.
- It assumes firms expect market-year effects to remain fixed even though industry-wide radio listening trends were changing.
- It assumes serially independent private payoff shocks.

Interpretation: The model is rich but still a simplification. The credibility of the counterfactual rests on whether the omitted margins are less important than format repositioning incentives.

7 Empirical results

7.1 Listener demand estimates

- Format tastes vary strongly by demographics.
- Music formats attract many listeners, including demographics valuable to advertisers.
- Station quality is persistent, with an estimated persistence parameter around 0.84.
- Switching formats is estimated to reduce unobserved quality modestly.
- The model matches the distribution of share changes for switching and non-switching stations reasonably well.

Interpretation: The demand model supports the idea that avoiding fees by switching away from music entails a real audience cost.

7.2 Revenue-function estimates

- Female listeners and listeners aged 25–49 are more valuable to advertisers.
- Young and older listeners are worth less than the 25–49 benchmark.
- Black and Hispanic listeners are estimated to be less valuable to advertisers than comparable white listeners in the reduced-form revenue function.
- Switching stations appear to discount commercials while building new relationships, so format switching reduces revenue in the short run.

Interpretation: These revenue differences matter because a station’s optimal format depends not only on audience size but also on the demographic composition of that audience.

7.3 Dynamic parameter estimates

- Repositioning costs are higher in larger markets.
- Moving from active to Dark is costly, especially in large markets.
- Economies of scope appear modest.
- Payoff shocks are larger in large markets.
- Forward-simulation estimates provide broad support for the parameter magnitudes used in the counterfactual.

Interpretation: The model estimates that format changes are costly enough to generate gradual adjustment, but not so costly that stations never move after a large policy shock.

8 Counterfactual: performance-rights fees

8.1 Implementation

- Fees are imposed unexpectedly in Fall 2004.

- Fees are calculated as a percentage of advertising revenues for music stations.
- The main cases are 10% and 20% of music-station revenues.
- Markets are simulated forward for up to 40 periods, corresponding to 20 years.
- The paper reports both the number of music stations and music/nonmusic listening shares.

Interpretation: The counterfactual is dynamic because firms can choose when, not just whether, to switch formats.

8.2 Main counterfactual results

- Without fees, the number of music stations remains approximately constant in the simulation.
- A 10% fee reduces the number of music stations by about 4.6% after one year and about 9.4% in the long run.
- A 20% fee produces a larger decline: about 7.1% after one year and about 20% in the long run.
- Music listening falls by less than the number of music stations because remaining music stations gain listeners.
- Nonmusic listening rises as some stations switch away from music.

Interpretation: The number of music stations falls more than music listening because the market reallocates listeners toward surviving music stations.

8.3 Role of heterogeneity and switching costs

- Taste heterogeneity limits switching because many listeners, including advertiser-valued listeners, prefer music.
- High repositioning costs slow adjustment.
- Increasing the cost or payoff-shock variance changes the transition path, but the qualitative conclusion remains: music stations decline under fees.

Interpretation: The model's richness matters because a simpler model with homogeneous listeners or frictionless switching could predict much larger or much faster changes.

9 Tables and figures

9.1 Table I and Table II: formats and summary statistics

Read:

- Table I shows the aggregation of BIAfn detailed formats into seven active categories plus Dark.
- Music formats include AC/CHR, Rock/AOR/Classic Rock, Country, and Urban.
- News/Talk and Spanish formats are treated as nonmusic in the counterfactual.
- Table II summarizes 714 market-periods and 16,481 station-periods.
- The median market has 18 stations; average annual station advertising revenue is about \$1.5 million.

Economic message: These tables describe the empirical environment: medium-sized differentiated-product markets with many stations and substantial variation in formats, listeners, and revenues.

TABLE I
FORMAT AGGREGATION^a

Aggregated Format (* = music format)	BIAFN Format Categories	Number of Station-Qtys		Mean Audience Share	% of Listeners in Demographic				
		FM	AM		Male	12-24	25-49	Black	Hisp.
1. AC/CHR*	Adult Contemporary, Rock, AOR/Classic Rock	3,299	47	5.3%	39.1	28.4	49.8	13.1	15.4
2. Rock, AOR/Classic Rock*	Contemporary Hit Radio	2,597	6	4.7%	70.5	25.6	63.0	2.3	8.9
3. Country*	Country	1,799	223	6.3%	47.4	15.1	44.2	2.4	5.0
4. Urban*	Urban, Gospel	1,172	609	4.5%	43.5	29.3	50.2	83.6	5.0
5. News/Talk	News, Talk, Sports	267	2,656	3.2%	60.7	3.7	38.1	7.5	5.1
6. Spanish	Spanish-language	562	460	2.2%	51.4	20.0	59.9	0.8	91.3
7. Other Programming	Oldies, Easy Listening, Variety, Classical, Jazz, Big Band, Religious (non-Gospel)	1,873	911	3.2%	47.1	6.8	36.4	9.9	10.2

^aDemographic percentages based on Arbitron's *Radio Today* publications 2003–2006 and author's calculations.

Table I: format aggregation

TABLE II
SUMMARY STATISTICS

Variable	Observations (market-periods)	Mean	Std. Dev.	Min.	Ma
Market Characteristics					
Number of stations in market	714	23.2	5.8	11	38
Number of different station owners in market	714	9.4	3.2	3	18
Population (12 and above, in millions)	714	0.73	0.75	0.06	3.4
Proportion of population black	714	0.11	0.12	0.002	0.4
Proportion of population Hispanic	714	0.09	0.10	0.005	0.4
Combined listening to sample stations (% of total market ^a)	714	35.0	3.3	22.8	45.1
BIAFN estimated annual market advertising revenues (2002–2005, \$ m.)	408	53.5	66.0	3.9	403.1
Station Characteristics (note: statistics exclude Dark stations)					
Market share	16,481	1.5%	1.3%	0.0%	10.1
BIAFN Estimated advertising revenues (station-year 2002–2004, \$ m.)	6,413	2.4	3.7	0.003	45.0
Dummy for station located outside market boundaries	16,481	0.06	0.24	0	1
Dummy for AM band	16,481	0.30	0.46	0	1
Proportion of market covered by signal (out of market stations excluded) ^b	15,464	0.79	0.36	0.002	1.0
Dummy for station that has imputed market share in at least one quarter	16,481	0.14	0.35	0	1
Dummy for station that switches formats before next quarter	14,120	0.032	0.18	0	1
Market share of a switching station	454	1.0%	0.9%	0.0%	6.1

^aThe total market includes the outside good of not listening to radio and listening to noncommercial and commercial nonsample stations. Market definition allows for individual to spend up to 6 hours listening to the radio between 6am and midnight.
^bSignal coverage is defined relative to the market population and I cap it at 1.1 to address outliers that appeared to distort the demand estimates.

Table II: summary statistics

9.2 Tables III–IV and Figure 1: demand estimates and fit

Read:

- Table III reports estimated format taste parameters by demographic group.
- The estimates show strong format-demographic sorting.
- Table IV shows station-quality parameters: AM-format penalties, signal coverage effects, quality persistence, and the quality effect of format switching.
- Figure 1 compares the distribution of observed share changes with model predictions for switching and non-switching stations.

Economic message: The demand system gives the counterfactual its discipline: stations switch only if saving fees outweighs audience and revenue losses.

TABLE III
ESTIMATES OF FORMAT TASTE PARAMETERS^a

	Mean Tastes	γ^{FM}	Age 25–49	Age 50 plus	Female	Black	Hisp.
Radio Listening	–15.8753 (0.9224)	16.7749 (1.3946)	2.1404 (0.2046)	7.7839 (0.7247)	2.8197 (0.2098)	3.6171 (0.3303)	2.8
Format Interactions (AC/CHR excluded)							
Rock	0.6575 (0.0580)	–	0.2813 (0.0087)	–0.4054 (0.0158)	–1.2581 (0.0043)	–1.8814 (0.0292)	–0.0
Country	–0.1187 (0.0646)	–	0.4847 (0.0095)	1.3039 (0.0175)	–0.4428 (0.0046)	–1.9726 (0.0530)	–1.0
Urban	–1.2040 (0.0832)	–	–0.7373 (0.0393)	–1.0188 (0.0497)	–0.3996 (0.0086)	3.9575 (0.0406)	0.6
News/Talk	–1.2918 (0.1443)	–	1.6979 (0.0080)	3.1485 (0.0144)	–1.1171 (0.0049)	–0.7935 (0.0275)	–1.0
Other Programming	–0.9883 (0.0650)	–	1.0958 (0.0079)	2.4600 (0.0150)	–0.5384 (0.0049)	–0.4204 (0.0275)	–0.0
Spanish	–2.7945 (0.1955)	–	1.0300 (0.0264)	1.1111 (0.0506)	–0.3649 (0.0163)	–0.5138 (0.1519)	3.9

^a16,481 observations, GMM objective function 2.90e–12, standard errors in parentheses. Mean tastes will reflect valuations of a white male aged 12–24.

Table III: format taste parameters

ESTIMATES OF STATION QUALITY PARAMETERS^a

AM * AC/CHR or Rock	–0.7781 (0.3174)
AM * Country	–0.4538 (0.1776)
AM * Urban	–0.3523 (0.1296)
AM * News/Talk	–0.0806 (0.1658)
AM * Other	–0.3811 (0.1196)
AM * Spanish	–0.2714 (0.1593)
Signal Coverage (for stations located in the market)	1.5938 (0.1004)
FM * Signal Coverage	0.6057 (0.1068)
Small Station Dummy (shares imputed for some quarters)	–0.4277 (0.0580)
Out of Market Dummy	–0.5082 (0.0834)
Transition Process for Unobserved Quality ρ^{FM}	0.8421 (0.0058)
σ_{FM}	0.3132 (0.0020)
Effect of Format Switch on Unobserved Quality	–0.0501 (0.0103)

^a16,481 observations, GMM objective function 2.90e–12. Time coefficients not reported. Std. errors in parentheses. AM * AC/CHR and *Rock combined due to small number of observations.

Table IV: station quality parameters

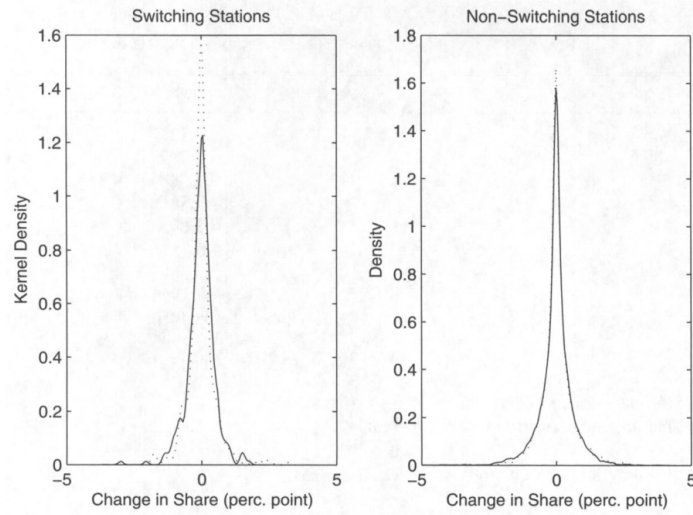


FIGURE 1.—Distribution of share changes: data (solid line), model prediction (dashed line, e simulation draw).

9.3 Tables V–VII: revenue and dynamic estimates

Read:

- Table V shows that listener demographics have strong effects on advertising value.
- Table VI reports dynamic estimates, including repositioning costs, dark-format costs, economies of scope, and payoff-shock scales.
- Table VII compares forward-simulation estimates and moment inequality bounds.

Economic message: The revenue and dynamic estimates jointly determine how much a station loses by switching away from music and how costly it is to do so.

TABLE V
PARAMETER ESTIMATES FOR THE REVENUE FUNCTION^a

	(1)	(2)
<i>Demographics</i>		
Female	0.1797 (0.0368)	0.1917 (0.0374)
Age 12–24	−0.5811 (0.1075)	−0.5883 (0.1084)
Age 50+	−0.4531 (0.0577)	−0.4572 (0.0581)
Black	−0.1964 (0.0148)	−0.1961 (0.0155)
Hispanic	−0.1593 (0.0159)	−0.1596 (0.0159)
<i>Station Characteristics and Competition</i>		
Number of stations with same owner in format	—	0.0064 (0.0047)
Number of other stations in format	—	−0.0019 (0.0018)
Format switch in previous quarter	—	−0.1045 (0.0279)
R^2 (compared to a model with only market-year fixed effects)	0.3182	0.3208

^a4,483 annual station observations (observations with imputed shares excluded). Market-year coefficients not reported. Standard errors corrected for imprecision in the demand parameters.

to affect revenues. The demographic coefficients are similar in both speci-

Table V: revenue function

TABLE VII
ESTIMATES BASED ON FORWARD SIMULATION

	Repositioning Cost	Scope Economy	Scale of Payoff Shock
<i>Markets With Pop. > 1 m.</i>			
Moment inequality bounds (PPHI)	[2.194, 11.031]	[−0.081, 0.056]	[0.074, 1.729]
95% CI	[0.652, 13.877]	[−0.126, 0.102]	[0.011, 1.793]
BBL point estimate	18.668	0.337	3.771
std. error	(1.765)	(0.048)	(0.342)
proportion of inequalities violated	24.4%	—	—
<i>Markets With Pop. 0.25 m.–1 m.</i>			
Moment inequality bounds (PPHI)	[0.464, 3.421]	[−0.071, 0.031]	[0.015, 0.549]
95% CI	[0.232, 4.035]	[−0.082, 0.042]	[0, 0.568]
BBL point estimate	3.046	0.013	0.630
std. error	(0.190)	(0.006)	(0.043)
proportion of inequalities violated	10.0%	—	—
<i>Markets With Pop. < 0.25 m.</i>			
Moment inequality bounds (PPHI)	[0.230, 1.541]	[−0.022, 0.008]	[0.005, 0.251]
95% CI	[0.081, 1.690]	[−0.27, 0.014]	[0, 0.258]
BBL point estimate	2.148	0.011	0.455
std. error	(0.242)	(0.004)	(0.051)
% of BBL inequalities violated	22.4%	—	—

Table VII: forward-simulation estimates

Specification	(1) Modified Procedure P-Likelihood	(2) Modified Procedure Moments	(3) Iterated Procedure P-Likelihood
<i>Costs of Move to Active Format (\$ m.)</i>			
Markets with pop. 1 m. +	2.524 (0.380)	1.897 (0.441)	1.467 (0.409)
* Recent Format Switch	-0.001 (0.098)	-0.001 (0.110)	-0.000 (0.065)
Markets with pop. 0.25-1 m.	0.669 (0.157)	0.446 (0.167)	0.388 (0.102)
* Recent Format Switch	0.077 (0.082)	-0.213 (0.145)	-0.044 (0.067)
Markets with pop. < 0.25 m.	0.233 (0.099)	0.069 (0.048)	0.135 (0.063)
* Recent Format Switch	0.030 (0.023)	-0.0310 (0.105)	-0.017 (0.054)
* Revenue of Switching Station (all markets)	0.034 (0.168)	0.116 (0.211)	0.022 (0.118)
<i>Additional Cost of Move to Active From Dark (\$ m.)</i>			
Markets with pop. 1 m. +	-0.501 (0.220)	-0.445 (0.183)	-0.291 (0.095)
Markets with pop. 0.25-1 m.	-0.255 (0.166)	-0.495 (0.098)	-0.147 (0.069)
Markets with pop. < 0.25 m.	-0.061 (0.091)	-0.078 (0.032)	-0.035 (0.042)
<i>Cost of Moving From Active to Dark (\$ m.)</i>			
Markets with pop. 1 m. +	2.704 (0.744)	3.126 (0.964)	1.572 (0.503)
Markets with pop. 0.25-1 m.	0.636 (0.230)	1.636 (0.582)	0.369 (0.130)
Markets with pop. < 0.25 m.	0.199 (0.054)	0.654 (0.322)	0.115 (0.022)
Revenue of Switching Station	1.108 (1.330)	3.105 (1.506)	0.642 (1.003)
<i>Economies of Scope</i>			
Markets with pop. 1 m. +	0.134 (0.099)	-0.046 (0.072)	0.078 (0.075)
Markets with pop. 0.25-1 m.	0.026 (0.018)	0.102 (0.021)	0.015 (0.010)
Markets with pop. < 0.25 m.	-0.006 (0.012)	-0.014 (0.010)	-0.033 (0.029)
<i>Scale of ϵ's</i>			
Markets with pop. 1 m. +	0.517 (0.180)	0.408 (0.063)	0.300 (0.087)
Markets with pop. 0.25-1 m.	0.144 (0.055)	0.091 (0.037)	0.083 (0.026)
Markets with pop. < 0.25 m.	0.050 (0.014)	0.014 (0.007)	0.028 (0.009)

9.4 Tables VIII–IX: policy counterfactuals

Read:

- Table VIII reports the evolution of music stations and listening under 0%, 10%, and 20% performance-rights fees.
- A 10% fee produces a gradual decline in music stations; a 20% fee produces a larger decline.
- Music listening falls less than the number of music stations because remaining music stations gain audience share.
- Table IX explores sensitivity to taste heterogeneity, repositioning costs, and payoff-shock dispersion.

Economic message: Performance fees reduce music variety, but the magnitude depends on demand substitution, advertiser values, and dynamic repositioning frictions.

TABLE VIII
EVOLUTION OF THE NUMBER OF MUSIC STATIONS UNDER DIFFERENT PERFORMANCE RIGHTS FEES^a

Fee Level:	Music Stations			Music Listening			Nonmusic Listening		
	0%	10%	20%	0%	10%	20%	0%	10%	20%
Period prior to introduction (Fall 2004)	713	713	713	0.254	0.254	0.254	0.095	0.095	0.095
+1 period	714 (3.7)	701 (4.6)	693 (4.6)	0.254 (0.002)	0.250 (0.003)	0.246 (0.003)	0.094 (0.002)	0.097 (0.003)	0.101 (0.004)
+5 periods	715 (6.1)	682 (7.2)	626 (7.1)	0.253 (0.003)	0.243 (0.003)	0.228 (0.005)	0.094 (0.002)	0.103 (0.003)	0.110 (0.004)
+10 periods (5 years)	717 (8.0)	665 (9.8)	595 (9.8)	0.253 (0.004)	0.238 (0.004)	0.220 (0.006)	0.095 (0.003)	0.106 (0.003)	0.116 (0.005)
+20 periods (10 years)	716 (10.1)	651 (11.5)	582 (11.8)	0.252 (0.004)	0.237 (0.005)	0.220 (0.006)	0.096 (0.003)	0.106 (0.003)	0.117 (0.003)
+40 periods (20 years)	720 (10.4)	652 (10.9)	578 (12.0)	0.253 (0.004)	0.237 (0.005)	0.219 (0.005)	0.096 (0.003)	0.107 (0.004)	0.118 (0.004)

^aStandard deviations across 10 simulations in parentheses. Results based on sample of 51 markets. Music listening measured as the average combined market share of music stations across markets.

Table VIII: performance-fee counterfactual

TABLE IX
EVOLUTION OF THE NUMBER OF MUSIC STATIONS AND MUSIC AUDIENCES UNDER 10% FEES FOR DIFFERENT ASSUMPTIONS ON THE STRUCTURAL PARAMETERS RELATIVE TO EVOLUTION WITH NO FEES (FALL 2004 INDEXED TO 1)^a

Column:	True Parameters ("Base case")		Less Taste Heterogeneity		Higher Repositioning Costs		Higher Repositioning Costs and High σ^e	
	(1) Stations	(2) Music Audiences	(3) Stations	(4) Music Audiences	(5) Stations	(6) Music Audiences	(7) Stations	(8) Music Audiences
Fall 2004	1	1	1	1	1	1	1	1
+1 period	0.981 (0.006)	0.984 (0.005)	0.941 (0.010)	0.950 (0.012)	0.986 (0.002)	0.992 (0.003)	0.986 (0.004)	0.993 (0.002)
+5 periods	0.954 (0.010)	0.961 (0.009)	0.875 (0.014)	0.895 (0.013)	0.967 (0.006)	0.978 (0.006)	0.971 (0.008)	0.980 (0.008)
+10 periods (5 years)	0.930 (0.015)	0.941 (0.010)	0.810 (0.017)	0.851 (0.015)	0.940 (0.010)	0.959 (0.009)	0.952 (0.014)	0.964 (0.012)
+40 periods (20 years)	0.906 (0.014)	0.937 (0.010)	0.801 (0.018)	0.841 (0.016)	0.910 (0.011)	0.942 (0.009)	0.918 (0.016)	0.945 (0.009)

^aResults based on sample of 51 markets. Standard deviations across 10 simulations in parentheses. Music listening measured as the average combined market share of music stations across markets.

Table IX: sensitivity to structural assumptions

10 Short summary

- Sweetening studies how radio stations would reposition programming if music stations had to pay performance-rights fees.
- The paper estimates a dynamic oligopoly model with differentiated formats, demographic demand, station quality, multi-station ownership, and repositioning costs.
- The data cover 102 local radio markets from 2002 to 2005.
- The model uses value-function approximations to solve a large dynamic game.
- Listener demand estimates show strong demographic format preferences.
- Revenue estimates show that advertisers value listener demographics differently.
- Dynamic estimates imply significant repositioning costs and gradual adjustment.
- A 10% music performance fee reduces the number of music stations by about 4.6% within one year and by about 9.4% in the long run.
- A 20% fee produces a larger decline, around 20% in the long run.
- Music listening falls less than the number of music stations because remaining music stations absorb listeners.
- The paper's broader lesson is that policy effects on product variety require dynamic structural modelling when product repositioning is costly.

11 Conclusion

11.1 Core conclusion

Sweetening shows that performance-rights fees would reduce the number of music radio stations, but the decline would be gradual and limited by listener preferences and advertiser values. A fee equal to 10% of music-station revenue would cause meaningful repositioning away from music, but it would not eliminate music programming.

11.2 Economic intuition

Music stations face a tradeoff after the fee. Staying in music means paying the fee, but switching to nonmusic means losing listeners and possibly losing advertiser-valued demographics. Repositioning is also costly, so

even stations that eventually switch may delay. The equilibrium response therefore depends on demand substitution, advertising values, and dynamic repositioning costs.

11.3 Incremental takeaway

The paper demonstrates how a rich dynamic differentiated-products model can be used for policy design. Its incremental contribution is not only a prediction about radio performance fees, but also a framework for studying product-variety regulation when firms can reposition products over time.